Scalar Cyclic Finite Braid Images

Exact sigma–MGF coefficients in dimensions one through four

Computational verification note

17 July 2026

Abstract

We prove the finite-group sigma–MGF formula directly and test it on the scalar cyclic quotients of B_3. These examples isolate the scalar-center selection rule while permitting arbitrary representation dimension. Explicit Reynolds-projector ranks agree with both the elementwise power-trace formula and its conjugacy-class grouping in all tested dimensions $D = 1, 2, 3, 4$.

1 The representation and the proposed quantity

Fix $m \geq 2$, $D \geq 1$, and $\zeta = e^{2\pi i/m}$. Define

$$\rho(\sigma_1) = \rho(\sigma_2) = \zeta I_D.$$

The matrices are unitary and satisfy $\rho(\sigma_1)\rho(\sigma_2)\rho(\sigma_1) = \rho(\sigma_2)\rho(\sigma_1)\rho(\sigma_2)$, so this is a D-dimensional representation of B_3. Its image is C_m.

For a tuple $\tau = (\tau_1, \ldots, \tau_\ell)$ let

$$W_\tau = \bigotimes_{a=1}^{\ell} \mathrm{Sym}^{\tau_a}(\mathbb{C}^D), \qquad |\tau| = \sum_a \tau_a.$$

The coefficient under test is

$$[m_\tau]\mathcal{S}_{C_m}(\mathbf{t}), \qquad \mathcal{S}_{C_m}(\mathbf{t}) = \frac{1}{m}\sum_{j=0}^{m-1}\prod_{a\geq 1}\det(I - t_a\zeta^j I_D)^{-1}.$$

2 Proof of the formula and closed answer

For any finite $G \leq U(V)$, the coefficient of t^r in $\det(I - tg)^{-1}$ is the character of $\mathrm{Sym}^r V$ at g. Multiplying the factors and applying the Reynolds character formula gives

$$[m_\tau]\mathcal{S}_G = \frac{1}{|G|}\sum_{g\in G}\prod_a \mathrm{Tr}(g \mid \mathrm{Sym}^{\tau_a} V) = \dim(W_\tau^G).$$

Furthermore,

$$\det(I - tg)^{-1} = \exp\left(\sum_{k\geq 1}\frac{t^k}{k}\,\mathrm{Tr}(g^k)\right).$$

Grouping the last average into conjugacy classes yields the weight $|[c]|/|G| = 1/|C_G(c)|$, proving both proposed finite-image formulas.

In the present representation, $\zeta^j I_D$ acts on all of W_τ by $\zeta^{j|\tau|}$. Hence the Reynolds operator is either zero or identity:

$$\boxed{[m_\tau]\mathcal{S}_{C_m} = \mathbf{1}_{m||\tau|} \prod_a \binom{D+\tau_a-1}{\tau_a}.}$$

This proves the scalar selection rule and also determines the nonzero value, not merely its support.

3 Independent verification strategy

The program first closes the matrix group from the displayed generator and asserts order m, unitarity, and the braid relation. It then constructs an orthonormal occupation basis of every $\mathrm{Sym}^r \mathbb{C}^D$ inside $(\mathbb{C}^D)^{\otimes r}$. In that basis it forms

$$P_\tau = \frac{1}{m} \sum_{g \in C_m} \bigotimes_a \mathrm{Sym}^{\tau_a}(g)$$

and computes its rank. This is the direct target calculation.

The comparator never uses this projector. It obtains the complete homogeneous characters from Newton's recurrence

$$r h_r(g) = \sum_{k=1}^{r} \mathrm{Tr}(g^k) h_{r-k}(g), \qquad h_0 = 1,$$

then averages $\prod_a h_{\tau_a}(g)$ elementwise and, separately, by matrix conjugacy classes.

m	D	τ	$\dim W_\tau$	$\operatorname{rank} P_\tau$	formula
3	1	(1)	1	0	0
3	1	(3)	1	1	1
3	2	$(2,1)$	6	6	6
4	2	(4)	5	5	5
4	3	$(2,2)$	36	36	36
5	4	(5)	56	56	56

The maximum projector idempotence/Hermiticity error in this sweep is below 6×10^{-15}.

4 Reproduction and scope

From the repository root run

```
.venv/bin/python for_this_guy/finite_braid_images/
  cyclic_scalar/verification.py
.venv/bin/marimo edit for_this_guy/finite_braid_images/
  cyclic_scalar/notebook.py
```

after joining each wrapped command. The proof is valid for every m, D, τ; the table is a small numerical regression suite, not evidence substituted for the proof.